C0 - Reference-Only Baseline: Detailed Flow

Fixed-Phase Nominal Reference Replay

Load the complete FDDP nominal reference
$t_{ref}, q_{ref}, dq_{ref}, x_{ref}$
phase / contact metadata

Use the fixed reference clock
$t_{query} = t$
no contact-event phase correction

Interpolate nominal joint and pitch references
$q_{nom} = interp(t, q_{ref})$
$\theta_{ref} = interp(t, pitch_{ref})$

Apply no feedback correction
$\alpha_k = 0, \quad \Delta q_k = 0$

Form the position command
$q_{cmd} = q_{ref}(t)$

Position proxy and MuJoCo actuator application
position / rate limiting → data.ctrl → mj_step

Inject pitch disturbance and log baseline metrics
$e_\theta = \theta_{meas} - \theta_{ref}$
pitch error / height / contacts / joint tracking

Evidence boundary: C0 is a reference-only disturbance baseline, not an MPC controller. It provides the fixed-phase reference used to quantify C1 recovery improvement.